Question 2

[13.5 Marks | 30 Minutes]

(a) What is a distance measure? Explain **Euclidean distance** and **Manhattan distance (L1 norm). [4 marks]**

(b) Briefly describe **Mahalanobis distance** and explain when it is useful. **[3 marks]**

(c) Differentiate between **Cosine Similarity, Pearson Correlation, and Jaccard Similarity. [4 marks]**

(d) What is the **curse of dimensionality** in k-NN? **[2.5 marks]**

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Question 9 – K-NN Classification and Effect of k

(13.5 Marks | 30 Minutes)

Consider the following two-dimensional training data:

Point	X_1	X_2	Class
A	1	1	Red
B	2	2	Red
C	2	4	Blue
D	3	3	Blue
E	4	5	Blue
F	6	2	Red

An unknown point $P = (3,3)$ is to be classified using Euclidean distance.

(a) Calculate the Euclidean distance between P and every training point. [4 marks]

(b) Determine the predicted class of P for:

- $k = 1$
- $k = 3$
- $k = 5$

Show the selected neighbors and voting process in each case. [4.5 marks]

(c) Analyze how changing k affects the classification result in this example. Explain the relationship between small k, variance, stability, and noise sensitivity. [3 marks]

(d) If the training data are highly imbalanced toward the Blue class, explain why ordinary majority voting may produce a biased prediction and suggest a suitable modification. [2 marks]

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Question 10 — Simple Linear Regression from Data

[13.5 Marks | 30 Minutes]

Consider the following dataset:

$$(2, 2),\ (3, 4),\ (6, 4),\ (7, 9),\ (8, 10)$$

Assume a linear model:

$$f(x) = w_0 + w_1 x$$

(a) Construct the required equations for determining w_0 and w_1 using the least-squares/best-fit principle. **[4 marks]**

(b) Determine the values of w_0 and w_1 and hence obtain the best-fit regression equation. **[4 marks]**

(c) Use the resulting model to estimate Y when $X = 10$. **[2 marks]**

(d) Explain analytically why minimizing the sum of squared errors provides a criterion for finding the "best fit" line. **[3.5 marks]**

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